

BALIKESIR, Turkey

WMO: 171500

Lat: 39.617N

Lon: 27.917E

Elev: 335

StdP: 14.52

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.7	25.2	17.5	13.5	25.7	20.2	15.3	28.8	27.9	39.5	24.3	43.7	1.6	350	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.0	96.6	69.1	93.3	68.7	90.8	68.5	73.2	87.8	72.0	86.4	70.6	84.9	10.3	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.4	105.8	79.4	67.3	101.6	78.9	65.9	96.6	78.3	37.0	87.8	35.9	86.4	34.8	84.8	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.5	22.1	19.9	DB	16.3	103.3	6.7	3.2	11.5	105.6	7.5	107.5	3.8	109.3	-1.1	111.6
			WB	15.7	75.2	6.7	1.8	10.9	76.4	7.0	77.5	3.2	78.5	-1.6	79.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	40.6	43.4	47.9	55.2	64.1	72.3	77.2	77.4	70.0	60.9	50.7	43.3	(6)
	DBStd	14.44	7.60	7.65	6.77	6.31	5.45	4.82	3.66	3.92	5.08	6.38	7.47	7.93	(7)
	HDD50	975	303	207	122	22	0	0	0	0	0	5	79	237	(8)
	HDD65	3551	756	605	530	300	81	4	0	0	15	156	431	673	(9)
	CDD50	4142	12	22	58	176	439	670	842	850	601	343	100	29	(10)
	CDD65	1243	0	0	0	4	55	224	377	386	166	29	2	0	(11)
	CDH74	14447	0	0	13	146	895	2712	4405	4224	1716	317	18	1	(12)
	CDH80	6355	0	0	1	26	276	1182	2143	1990	658	78	1	0	(13)
(14) Wind	WSAvg	6.8	5.8	6.8	6.7	5.5	5.6	7.4	9.7	10.0	7.1	6.3	4.6	5.6	(14)
(15) Precipitation	PrecAvg	22.7	3.2	2.6	2.3	2.0	1.6	1.0	0.4	0.3	1.0	1.7	3.1	3.8	(15)
	PrecMax	39.2	10.8	7.9	5.5	5.8	4.0	3.8	2.1	2.0	4.7	8.0	6.8	11.8	(16)
	PrecMin	13.9	0.1	0.2	0.5	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.1	0.0	(17)
	PrecStd	5.2	2.2	1.6	1.2	1.1	1.0	1.0	0.5	0.5	1.1	1.5	1.6	2.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.3	68.2	75.2	83.9	91.3	98.9	102.2	100.3	96.4	87.5	76.9	67.0	(19)
		MCWB	53.4	54.6	57.5	62.6	65.3	68.7	69.7	69.0	66.5	64.7	60.9	56.0	(20)
	2%	DB	60.7	63.9	69.9	78.4	86.3	93.3	96.7	95.2	91.2	81.6	71.3	62.7	(21)
		MCWB	52.1	53.3	55.2	61.1	64.8	67.9	69.2	69.4	66.3	63.3	58.4	54.3	(22)
	5%	DB	56.9	60.2	66.2	73.7	82.7	89.9	93.2	92.4	86.6	77.2	66.8	59.2	(23)
		MCWB	50.2	51.5	53.9	59.0	63.5	67.2	69.1	69.7	65.7	61.9	56.9	52.7	(24)
	10%	DB	53.2	56.4	62.2	69.8	79.0	86.8	89.8	89.6	83.7	73.2	62.8	55.5	(25)
		MCWB	47.9	49.7	52.3	57.4	62.5	66.5	68.9	69.4	64.9	60.9	55.2	50.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.9	56.4	59.0	64.8	68.4	72.9	74.3	75.5	71.7	67.4	62.9	57.0	(27)
		MCDB	62.2	65.4	72.3	78.7	83.8	90.2	89.9	90.1	86.2	78.7	73.1	64.5	(28)
	2%	WB	52.3	54.0	56.4	62.1	66.5	70.7	72.8	73.7	69.4	65.3	60.0	54.8	(29)
		MCDB	59.4	62.4	67.3	75.7	82.1	86.2	88.3	87.5	82.6	76.3	68.5	61.5	(30)
	5%	WB	50.3	51.9	54.4	59.9	64.9	69.2	71.6	72.5	67.8	63.7	57.7	53.0	(31)
		MCDB	56.2	58.9	64.4	72.1	79.4	84.3	86.9	85.9	80.7	74.0	65.4	58.8	(32)
	10%	WB	48.2	50.0	52.4	57.7	63.4	67.8	70.3	71.1	66.4	62.0	55.6	50.9	(33)
		MCDB	52.8	56.1	61.1	68.5	76.6	82.9	85.5	84.2	79.8	71.1	62.2	55.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	16.9	20.6	23.4	25.8	26.4	25.0	24.1	25.2	21.5	19.6	14.3	(35)
		MCDBR	21.4	24.7	30.2	32.1	33.6	32.8	32.5	29.6	32.1	29.4	25.6	19.4	(36)
	5% WB	MCWBR	14.8	15.9	17.8	17.8	15.4	12.5	12.0	10.5	13.0	14.6	15.6	12.9	(37)
		MCWBR	19.9	22.4	27.1	29.4	29.2	27.6	25.8	24.3	25.6	24.0	23.0	18.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.363	0.397	0.438	0.482	0.452	0.424	0.422	0.430	0.419	0.408	0.376	0.364	(40)
		taud	2.321	2.197	2.090	1.998	2.152	2.277	2.287	2.260	2.277	2.281	2.318	2.327	(41)
	Ebn at Noon		249	255	257	253	264	272	271	266	260	249	241	238	(42)
		Edh at Noon	29	38	47	55	48	42	42	42	38	35	29	27	(43)
(44) All-Sky Solar Radiation	RadAvg		558	775	1191	1592	1997	2277	2410	2113	1608	1080	719	513	(44)
	RadStd		77	101	115	180	142	171	75	102	113	94	88	78	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.03	N/A	N/A	N/A	N/A	N/A	-99	-247	+214	+113	(46)
(47)	Regional (18 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	-193	+202	+104	(47)

Nomenclature: See separate page